

Noise Reduction and Sound Source Localization for Unmanned Ariel Vehicles (UAVs)

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Abstract—Unmanned Aerial Vehicles (UAVs) are becoming essential tools for public safety, particularly in high-risk situations such as emergency response and large-scale surveillance. Their mobility allows rapid deployment over areas that may be unsafe for human operators, offering real-time audio and visual intelligence to support decision making, in extreme scenarios such as gunshot detection and fire rescue. According to Wikipedia, Business Insider and National Law Enforcement Officers, on an average there were 630 gunshot events in every year from 2020 to 2024 and in this gunshot events every year 53 police officers were killed. UAVs can be deployed for massive gunshot scenarios to identify the source and direction of the shooter before sending the police officer to the site and risking their lives assessing the position and number of shooters. However, one of the major barriers to reliable UAV-based acoustic monitoring and source tracking is the strong ego-noise produced by propellers and motors. This noise, often louder than the sounds of interest, masks critical acoustic cues and reduces the effectiveness of detection and localization systems. Taking into consideration this challenge requires noise reduction strategies that can detect weak but meaningful signals, such as gunfire or distress calls, even under extremely low signal-to-noise conditions. By combining the advantage of deep learning with advanced audio processing techniques, UAVs can be transformed into effective auditory sensing platforms for critical event surveillance.

Index Terms—Deep neural network, recursive neural network, drone ego-noise, UAVs

I. INTRODUCTION AND MOTIVATION

Unmanned aerial vehicles (UAVs) are emerging as critical tools for law enforcement, emergency response and perimeter security. In active-shooter or high-risk scenarios, an aerial platform capable of autonomously detecting and localizing gunshots can provide fast situational awareness while keeping human responders at a safer distance. Unlike fixed acoustic sensor networks, a UAV can be rapidly deployed to previously unseen environments, navigate to vantage points that are difficult to reach from the ground and combine audio with onboard video and other sensors. For these missions, robust onboard gunshot detection and localization are key enablers, especially in cluttered urban environments where visual line-of-sight is often obstructed. Realizing this vision is technically challenging because a drone is an extremely hostile platform for acoustic sensing. Microphones mounted close to motors and propellers are exposed to intense ego-noise that can be

several tens of decibels stronger than the external sound sources of interest. This ego-noise exhibits a mixture of narrow band harmonic components, generated by rotating rotors, and broadband aerodynamic noise, generated by propellers cutting through the air; its spectrum and level fluctuate with motor speed and flight maneuvers, leading to highly nonstationary noise fields. As a result, classical single-channel denoising techniques and standard microphone-array beamforming often fail, especially when the SNR at the microphones falls below -10 to -15 dB or when the desired source lies in a similar direction to the rotors [1]. Deep learning has recently transformed drone audition by providing data-driven models that can explicitly learn the structure of ego-noise and its time-frequency overlap with speech and other sounds. Wang and Cavallaro showed that DNN-estimated ideal ratio masks can be used not only for single-channel drone ego-noise reduction, but also to drive multi-channel spatial filtering, yielding strong gains over traditional spectral subtraction, unsupervised noise estimators, and beamforming in extremely low-SNR conditions. Building on this, deep learning assisted sound source localization pipelines such as SRP-DNN and TFS-DNN combine single-channel DNN enhancement with multi-channel Steered Response Power and time-frequency spatial filtering, enabling robust DOA estimation of speech sources from flying drones in dynamic scenarios and short ($\approx 0.5s$) signal segments [1].

Despite this progress, most existing work targets continuous speech enhancement or generic sound source localization and does not explicitly address impulsive, high-consequence events such as gunshots. Gunshots differ from speech in several important ways such as they are short-duration, broadband impulses with strong high-frequency components; they may occur sparsely in time and they often coexist with multiple interfering sources, reverberation and overlapping rotor noise. A front end optimized for speech may therefore under-emphasize or even suppress the impulsive energy of gunfire, while localization pipelines tuned on quasi-stationary sources may not fully exploit the temporal sparsity and distinct spectral signatures of gunshots.

This paper focuses on gunshot detection and sound source localization from a quadrotor platform by explicitly integrating

ego-noise reduction, time-frequency masking and multi-channel spatial processing. We aim to bridge the gap between general-purpose drone audition and mission-oriented acoustic sensing for public safety. Concretely, our contributions are:

- 1) **A gunshot-aware ego-noise suppression front end** that leverages deep neural networks trained on UAV audio to estimate time-frequency masks which preserve broadband impulsive energy while attenuating rotor-dominated regions.
- 2) **A DNN-informed spatial localization module** that integrates the estimated masks into a multi-channel time-frequency spatial filtering framework, building on prior DNN-assisted methods for speech enhancement and localization on drones but adapting them to impulsive events and gunshot-like acoustics.
- 3) **An evaluation on UAV-mounted microphone data** combining public drone audition datasets and new recordings with realistic ego-noise, multiple ranges and different drone operating conditions, demonstrating improved gunshot detectability and more accurate DOA estimates compared to classical SRP-PHAT and mask-free time-frequency spatial filtering.

By explicitly tailoring the learning objectives and spatial filtering to impulsive sound events, this work brings drone audition one step closer to practical deployment in gunshot-detection and critical-incident response systems.

II. BACKGROUND AND RELATED WORK

A. UAV Ego-Noise and Classical Reduction Methods

The acoustic environment on a multi-rotor UAV is dominated by ego-noise from the motors and propellers. Measurements on micro-aerial vehicles show that this ego-noise consists of strong harmonic components aligned with the motor rotation frequencies and their multiples, superimposed on broadband turbulence noise from the propeller blades. Variations in rotor speed during hovering, acceleration, and maneuvering lead to time-varying fundamental frequencies and spectral envelopes, making the noise highly nonstationary. Classical single-channel noise reduction algorithms, such as spectral subtraction and minimum-mean-square-error (MMSE) estimators, assume slowly varying noise statistics and thus struggle when applied to UAV recordings [2]. Multi-channel beamforming and microphone-array techniques can, in principle, steer nulls towards rotor directions, but in practice their performance is degraded when the desired source lies close to the rotor DOAs or when the SNR at the microphones is extremely low.

To address these limitations, time-frequency spatial filtering methods have been proposed that exploit the sparsity of speech and ego-noise in the joint time-frequency domain. In these approaches, the instantaneous DOA of the sound at each time-frequency bin is estimated, and bins whose DOA is consistent with the target direction are used to build spatial filters, such as multi-channel Wiener filters, that suppress ego-noise while

preserving the target signal [3]. Although these time-frequency spatial filters offer improved robustness in low-SNR scenarios, their correlation-matrix estimates are still contaminated when the target source and the ego-noise share similar directions, leaving residual rotor noise in the enhanced signal.

B. Deep-Learning-Based Ego-Noise Suppression on Drones

Deep neural networks (DNNs) have been widely adopted for speech enhancement in stationary and nonstationary noise and have recently been adapted to the drone audition setting. Wang and Cavallaro proposed a DNN that estimates ideal ratio masks in the time-frequency domain, using fully connected architectures or more compact convolutional networks, and showed that these masks can substantially improve perceived speech quality and SNR for UAV recordings compared to traditional unsupervised algorithms [4]. Their framework includes single-channel ego-noise reduction, DNN-assisted beamforming, and a DNN-assisted time-frequency spatial filtering method (DNN-TF) that integrates the DNN-inferred noise dominance probabilities into the spatial filter design. DNN-TF, in particular, achieves strong performance at SNRs below -15 dB by excluding noise-dominated time-frequency bins from the target correlation matrix, especially when the target DOA is close to rotor directions [4]. Complementary work has investigated architectures specialized for UAV noise characteristics. Dilated convolutional neural networks, such as SMOlNet, operate in the complex spectrogram domain with stacked dilated convolution layers to aggregate information across frequency while maintaining a compact model size suitable for embedded processors. These models have been shown to provide efficient suppression of ego-noise at very low input SNRs with reduced computational cost compared to fully connected large networks. Such architectures are attractive building blocks for UAV-mounted gunshot detection because they can run in real time on resource-constrained hardware.

C. Deep-Learning-Assisted Sound Source Localization on Drones

In parallel, deep learning has been integrated into sound source localization (SSL) pipelines for flying drones. Traditional spatial likelihood methods such as SRP-PHAT and high-resolution algorithms like MUSIC perform well on ground robots but degrade on drones because they treat all time-frequency bins equally and are highly sensitive to strong ego-noise. To overcome this, Wang and Cavallaro proposed a deep learning assisted SSL framework that uses a single-channel DNN to estimate a soft ratio mask, then applies this mask as a time-frequency weight inside multi-channel SRP and time-frequency spatial filtering algorithms.

The resulting SRP-DNN and TFS-DNN pipelines exploit both spectral and spatial information. SRP-DNN multiplies the SRP-PHAT contribution of each time-frequency bin by the DNN-estimated speech-presence probability, yielding spatial likelihood maps that remain reliable at lower SNRs than SRP-PHAT. TFS-DNN combines the instantaneous DOA estimates from time-frequency spatial filtering with the DNN mask to

compute a confidence measure that a given bin originates from a candidate direction; this confidence then guides the construction of spatial filters whose outputs are analyzed via kurtosis to build a spatial likelihood map. Experiments on datasets such as AS, AVQ, and DREGON, which include static and moving sources as well as flying drones, show that TFS-DNN can localize speech sources accurately even when SNR is as low as -20 dB and only short (≈ 0.5 s) audio segments are available [4].

D. Gap Towards Gunshot Detection and Localization

While the above methods demonstrate that deep-learning-assisted time-frequency processing and spatial filtering can make drone audition viable at extremely low SNRs, the application space has largely focused on continuous speech or generic broadband sounds. Existing architectures are not explicitly tuned to preserve the fine-time structure and broadband spectral peaks of gunshots and similar impulsive events, and the evaluation datasets typically do not include firearm discharges or firearm-like impulsive sources. Moreover, most prior work treats enhancement and localization separately and does not integrate the detection of specific acoustic events into the localization pipeline.

This motivates a dedicated study of gunshot detection and localization on drones that leverages the strengths of DNN-based ego-noise suppression and time-frequency spatial filtering while explicitly modeling the temporal and spectral characteristics of gunshots. The present work situates itself at this intersection, aiming to design masks, front ends, and spatial filters that are “gunshot-aware” and to evaluate them under conditions representative of real public-safety deployments.

III. DATASET DESCRIPTION

The experiments in this work are based on the Gunshot Audio Forensics Dataset collected by Cadre Forensics under U.S. NIJ grant 2016-DN-BX-0183 [5].

The original dataset was recorded in rural Arizona and contains controlled test fires from approximately twenty firearms, captured at multiple distances and azimuth angles using four recording devices: a Zoom H4n handheld recorder, an iPhone 7, a Samsung Galaxy S7, and a body-worn camera. Each firearm was recorded from around twenty different geometries, and each shot was captured simultaneously on all devices, yielding on the order of ten thousand individual gunshot recordings.

In this project, we use a subset of this corpus consisting of eighteen firearm classes that cover common pistols, revolvers, carbines, and rifles across calibers such as .22 LR, .380 Auto, 9 mm, .40 S&W, .45 Auto, 5.56 mm, 7.62 mm, and .308 Winchester.

All audio files are converted to single-channel, 32 kHz pulse-code modulation (PCM) waveforms and trimmed so that each file contains a single shot centered around the muzzle blast. The original data are organized according to a consistent

naming convention that encodes device, geometry, and shot index and the geometries span azimuth angles from -30° to 180° and distances from 0.5 m to 40 m.

For this Phase 1 study, we focus on recordings from the handheld devices like Zoom, iPhone, Samsung and treat each firearm recording as an independent labeled sample in an 18-class supervised classification problem. The dataset is split into disjoint training, validation and test partitions at the recording level so that the same underlying gunshot does not appear in more than one split.

IV. METHODOLOGY

A. Model Architecture

To establish a baseline for gunshot recognition that will later be embedded in the UAV pipeline, we train two complementary deep learning models:

1) **Convolutional Neural Network (CNN) classifier:** The first model is a 2D CNN that operates on log-mel spectrogram “images” of each gunshot.

A typical architecture consists of a stack of convolutional blocks, each containing:

- a Conv2D layer with small 3x3 kernels,
- batch normalization,
- a rectified linear unit (ReLU) non-linearity, and
- 2x2 max-pooling to gradually reduce the time–frequency resolution.

After six such blocks, the feature maps are flattened and passed through two fully connected layers with dropout regularization. The final layer is a fully connected layer with 18 output units and a softmax activation, which produces a probability distribution over the firearm classes. This CNN is designed to capture local time-frequency patterns in the muzzle blast and early reflections that are characteristic of each firearm type.

2) **LSTM-based recurrent neural network (LSTM-RNN):** The second model treats each gunshot as a short time series of frame-wise feature vectors. The input to the LSTM is a sequence $\{x_t\}_{t=1}^T$, where x_t is a mel-spectral feature vector for frame t . We use a stacked BiLSTM with two recurrent layers, which can model the temporal evolution of the gunshot energy and reflections. The final hidden state is fed into a fully connected layer and a softmax output with 18 units.

In Phase 1, both architectures are trained and evaluated on exactly the same 18-class dataset, allowing a direct comparison between image-based (CNN) and sequence-based (LSTM-RNN) representations for gunshot recognition.

B. Feature Engineering

For both models, we start from the raw waveform and compute time–frequency representations that are standard in audio and speech processing and have also been used in prior gunshot analysis work.

- Each recording is peak-normalized and The waveform is then segmented around the gunshot so that the fixed-length window includes the muzzle blast, early reflections, and a short amount of pre-shot background. This

is analogous to prior work that models a central window containing the shot plus a context window before and after the event.

- We compute a short-time Fourier transform (STFT) with a frame length on the order of tens of milliseconds and then map the magnitude spectra to a mel-frequency scale with k mel bands (for example, $K=64$). The mel spectrogram is converted to a log scale to emphasize relative energy differences and to compress the large dynamic range characteristic of gunshots. Each recording is therefore represented by a matrix of size $K \times T$, where T is the number of time frames.
- For the CNN model, the log-mel spectrogram matrix is treated as a 2D image with one channel. We cropped and zero-pad in the time dimension so that all spectrograms have the same number of frames. The resulting tensor of shape $(1, K, T)$ is used directly as the CNN input.
- For the LSTM-RNN, each time frame corresponds to a feature vector $x_t \in \mathbb{R}^K$. The sequence $\{x_t\}_{t=1}^T$ is optionally standardized to zero mean and unit variance per feature dimension. To handle variable-length events, we either:
 - limit the sequence to a maximum number of frames (truncation).
 - pad shorter sequences with zeros and use a mask during training.

This representation allows the LSTM to learn how the energy in each mel band evolves over the duration of the gunshot.

C. Training Setup

Training is formulated as a supervised 18-class classification problem.

- The dataset is partitioned into training, validation, and test sets with no overlap in recordings between splits. A typical split reserves around 80% of the data for training and validation the remainder for testing.
- Both models are trained using categorical cross-entropy loss and optimized with the Adam optimizer. Mini-batch training is used and the learning rate is tuned empirically within a small range (for example, 10^{-3} to 5×10^{-4}).
- During training we monitor overall accuracy on the validation set and, after convergence, we compute test-set accuracy, per-class precision, recall, and F1-score, along with confusion matrices for both the CNN and LSTM models. These metrics reveal which firearm types are consistently recognized and which are confused, providing a baseline for the Phase 2 work where the same classifiers will be embedded into a UAV platform and exposed to strong ego-noise.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

The proposed CNN-based gunshot classifier is evaluated on the 18-class subset of the Cadre Forensics gunshot dataset

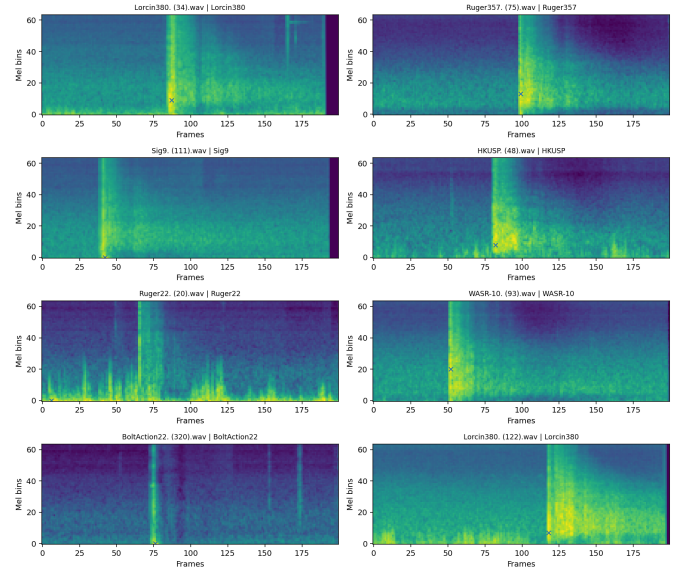


Fig. 1. Example log-mel spectrogram inputs for several gunshot recordings from different firearm classes. Each panel shows one recording; brighter colors indicate higher energy.

described in Section III. The recordings are split into disjoint training, validation, and test sets so that no waveform appears in more than one split. All models are trained on log-mel spectrogram inputs with the architecture and feature extraction pipeline described in Section IV. During training we monitor validation accuracy as a function of iteration and final performance is reported on the held-out test set via a confusion matrix and per-class analysis.

B. Qualitative Analysis of Log-Mel Inputs

Figure 1 shows example log-mel spectrograms for eight gunshot recordings from different firearm classes. In each case, the impulsive muzzle blast appears as a bright, nearly vertical ridge in the time-frequency plane, followed by a short decay and lower-energy reverberation. Although all classes share this general impulsive pattern, the detailed structure of the spectrum for example, relative energy across mel bins and the temporal decay varies from weapon to weapon and with recording conditions. These differences provide the CNN with discriminative cues to separate firearm types despite their similar overall shape.

C. Embedding Space Visualization

To better understand how the trained CNN separates classes, we apply t-distributed stochastic neighbor embedding (t-SNE) to the penultimate-layer embeddings of the training samples. The resulting two-dimensional projection is shown in Fig. 2. Each point corresponds to one recording, and colors indicate firearm classes.

The t-SNE plot reveals that the CNN learns a highly structured embedding space, most classes form compact, well-separated clusters. For example, several pistol and revolver classes occupy distinct regions with little overlap, indicating

t-SNE of embeddings (train trainIter050000, fold=1, perp=35.0, tsnelter=800, metric

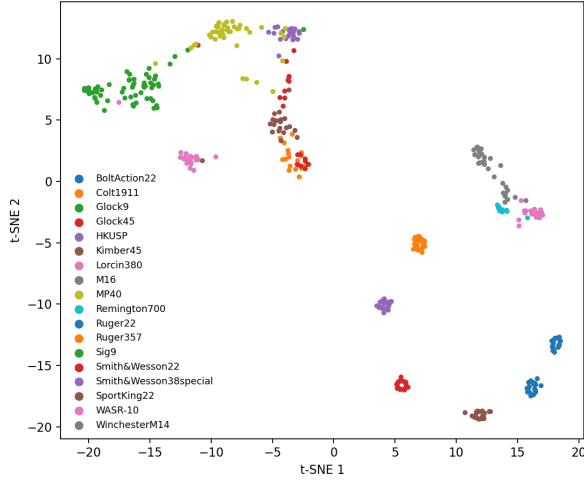


Fig. 2. t-SNE visualization of CNN embeddings for the 18 firearm classes. Each point represents one recording; colors denote classes. Most classes form compact, well-separated clusters, indicating good discriminability in the learned feature space.

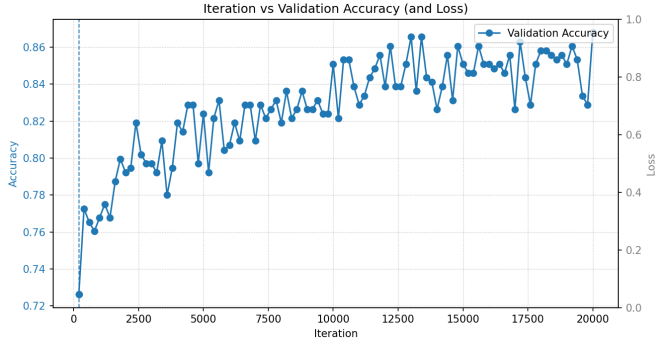


Fig. 3. Validation accuracy as a function of training iteration for the CNN model. Accuracy improves steadily and stabilizes around 86% near 20 000 iterations.

that the network has learned class-specific representations. A few rifle-type classes appear closer together, which foreshadows the confusions observed in the confusion matrix (Section V-D). Overall, the embedding visualization confirms that the CNN is not merely memorizing individual examples but is organizing recordings into meaningful, class-dependent manifolds.

D. Classification Performance

Figure 3 shows the evolution of validation accuracy over training iterations. The accuracy rises quickly from about 73% in the early iterations to around 80% within the first few thousand iterations, and then continues to improve more gradually, stabilizing near 86% by 20 000 iterations. The absence of a sharp drop in validation accuracy suggests that the chosen architecture and regularization are sufficient to avoid severe overfitting for this dataset size.

Final test-set performance is summarized by the confusion matrix in Fig. 4. The overall test accuracy is 75.56%, indi-

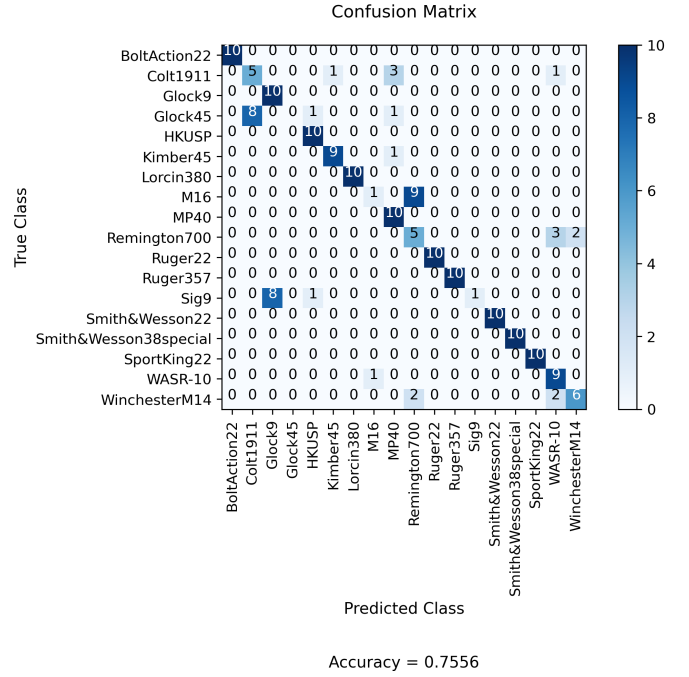


Fig. 4. Confusion matrix for the 18-class CNN gunshot classifier on the held-out test set. The diagonal entries correspond to correctly classified samples; the overall accuracy is 75.56%. Most classes achieve high accuracy, with a few confusions between acoustically similar rifle classes.

cating that the model correctly identifies roughly three out of four gunshot recordings across the 18 classes. Most classes exhibit strong performance, many diagonal entries reach 9 or 10 out of 10, meaning that both precision and recall are close to 1.0 for those firearm types. This is consistent with the tight, isolated clusters observed in the t-SNE embedding. The confusion matrix also highlights where the model struggles. A small number of rifle-type classes show noticeable off-diagonal entries, with some recordings being misclassified as acoustically similar weapons. These confusions suggest that certain firearms share very similar spectral signatures at the microphone, especially when recorded under comparable geometries. In future work, incorporating additional cues such as multi-channel spatial information or longer context windows may help disambiguate these hard cases.

In summary, the CNN trained on log-mel spectrograms learns discriminative embeddings that separate most firearm classes cleanly, achieves validation accuracy near 86%, and attains 75.56% test accuracy. These results provide a strong baseline for the next phase of the project, where the same architecture will be extended to operate under UAV ego-noise and integrated into a drone-based gunshot detection and localization system.

VI. RESULTS FOR LSTM-BASED GUNSHOT CLASSIFICATION

A. Qualitative Analysis of Log-Mel Sequences

To verify that the input features capture the temporal structure of the gunshots, we inspected the log-mel spectrogram

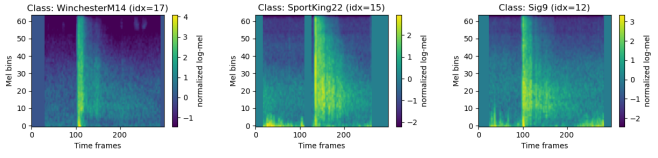


Fig. 5. Example log-mel spectrogram sequences used as input to the BiLSTM model for three firearm classes (WinchesterM14, SportKing22, Sig9). The bright vertical ridge corresponds to the muzzle blast, followed by lower-energy reverberation.

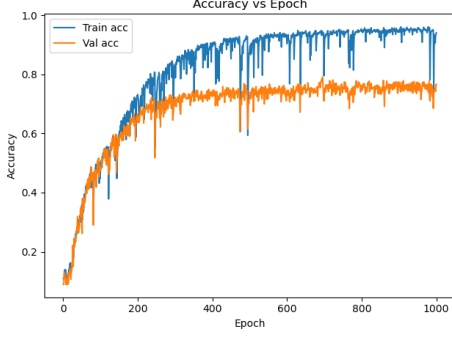


Fig. 6. Training and validation accuracy versus epoch for the BiLSTM gunshot classifier. Training accuracy approaches 1.0, while validation accuracy stabilizes around 75%, indicating good but not perfect generalization to unseen data.

sequences fed to the BiLSTM. Figure 5 shows examples for three validation recordings from different firearm classes (WinchesterM14, SportKing22, and Sig9). In each case, the muzzle blast appears as a high-energy vertical band followed by a decaying tail and weaker reverberation. Although all classes share this impulsive pattern, the detailed energy distribution across mel bins and the decay shape differ between weapons, providing discriminative temporal cues for the recurrent model.

B. Training and Validation Accuracy

Figure 6 shows the evolution of training and validation accuracy over 1000 epochs. The training accuracy increases rapidly during the first 200 epochs and then gradually approaches values close to 1.0, indicating that the BiLSTM has sufficient capacity to fit the training data. The validation accuracy follows a similar trend but saturates around 0.75 after approximately 400–500 epochs. The gap between training and validation curves suggests some degree of overfitting, but the validation curve remains stable rather than degrading, indicating that the model generalizes reasonably well to unseen recordings.

C. Validation Confusion Matrix

The detailed class-wise behavior of the BiLSTM is illustrated by the validation confusion matrix in Fig. 7, which reports both raw counts and row-normalized percentages. The matrix is strongly diagonal, confirming that most recordings are assigned to the correct firearm class. Several classes

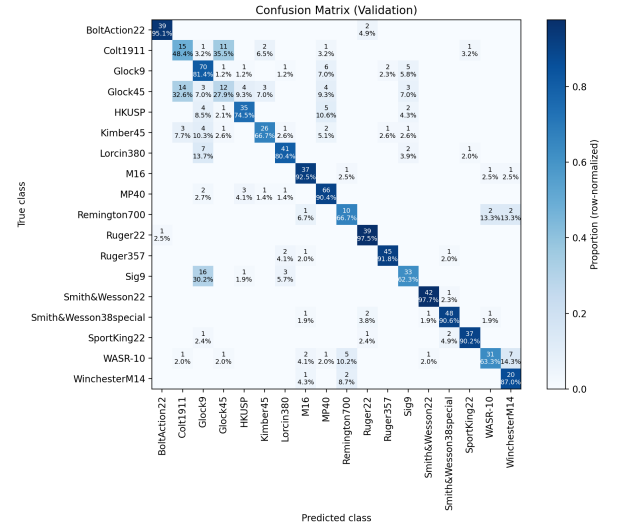


Fig. 7. Validation confusion matrix for the BiLSTM gunshot classifier. The diagonal entries (counts and percentages) show that most firearm classes are recognized with high accuracy, while a few classes exhibit notable confusion with acoustically similar weapons.

achieve very high validation recall (with diagonal entries exceeding 90%), indicating that the model recognizes these weapons reliably. A few classes exhibit lower diagonal percentages and noticeable off-diagonal entries, reflecting confusions between acoustically similar firearms and classes with fewer training examples. Overall, the model maintains robust performance across the majority of the 18 classes while exposing a small set of harder cases that will be targeted in future work.

D. Embedding Space Visualization

To analyze how the BiLSTM organizes gunshot recordings in its internal representation, we extract the final hidden-state embeddings for all validation samples and project them into two dimensions using t-distributed stochastic neighbor embedding (t-SNE). The resulting plot is shown in Fig. 8. Each point corresponds to a single recording and is colored according to its ground-truth class. The t-SNE visualization reveals that the BiLSTM learns a structured, class-dependent embedding space. Most classes form compact, well-separated clusters with relatively little overlap, which is consistent with the strong diagonal structure of the confusion matrix. Classes that are more frequently confused tend to lie closer together or exhibit partial overlap in the embedding space, indicating that their acoustic signatures are harder to distinguish. These results suggest that the recurrent model is capturing meaningful, class-specific temporal patterns in the gunshot waveforms rather than memorizing individual examples.

In summary, the BiLSTM-based classifier trained on log-mel sequences achieves validation accuracy of roughly 75%, with strong performance on most firearm classes and clear, class-specific structure in the learned embedding space. These results provide a complementary sequence-based baseline to

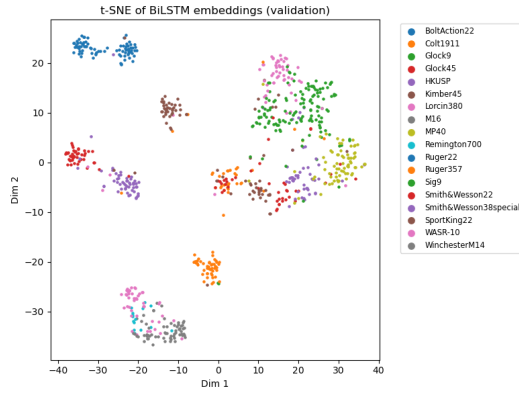


Fig. 8. t-SNE visualization of BiLSTM embeddings for validation recordings. Each point represents one recording, colored by firearm class. Most classes form distinct clusters, indicating that the learned representation is highly discriminative.

the CNN model and will serve as a foundation for the next phase, where both architectures will be extended to operate under UAV ego-noise and integrated into the drone-based gunshot detection pipeline.

VII. DISCUSSION AND LIMITATIONS

The Phase 1 results show that both the CNN and BiLSTM architectures can learn discriminative representations of firearm-specific gunshot signatures from log-mel features. The CNN trained on spectrogram “images” achieves validation accuracy near 86% and test accuracy of 75.56%, while the BiLSTM operating on frame-wise sequences reaches validation accuracy of roughly 75%. Taken together with the t-SNE visualizations and confusion matrices, these results suggest that the models are capturing meaningful spectral and temporal patterns rather than simply memorizing individual recordings. From a modeling perspective, the CNN tends to perform slightly better in terms of overall accuracy and produces very compact clusters in the embedding space, especially for pistol and revolver classes. This indicates that local time-frequency patches are highly informative for differentiating many firearms. The BiLSTM, on the other hand, emphasizes the temporal evolution of the muzzle blast and reverberation. It achieves strong performance on most classes and yields a structured embedding space, but shows a somewhat larger gap between training and validation accuracy, indicating a higher risk of overfitting.

This Phase 1 study has several limitations. First, all experiments are conducted on clean ground-based recordings from the Cadre Forensics dataset. No UAV ego-noise is present in the training or evaluation data, so the current models do not yet address the core challenge of low-SNR operation on a flying platform. Second, the models are trained and evaluated on a fixed subset of 18 firearm classes; real-world deployments may encounter weapons or calibers not represented in the training set, requiring strategies for open-set recognition or unknown-class detection. Third, the current work focuses on shot-level classification only and does not address sound

source localization or multi-shot event detection over longer time windows. Finally, all processing is performed offline on a GPU. Embedded deployment on resource-constrained hardware such as an NVIDIA Jetson module will require careful optimization of model size, memory footprint, and latency.

VIII. CONCLUSION

This paper presented a Phase 1 study toward UAV-based gunshot detection that focuses on establishing strong baseline classifiers using clean firearm recordings. Using an 18-class subset of the Cadre Forensics gunshot dataset, we developed two complementary deep learning models: a 2D CNN operating on log-mel spectrogram images and a BiLSTM that processes frame-wise log-mel sequences. The CNN achieves an overall test accuracy of 75.56% and produces well-separated class clusters in the embedding space, while the BiLSTM attains validation accuracy of approximately 75% with similarly structured, class-dependent embeddings. These results demonstrate that both architectures can reliably distinguish between a diverse set of firearms based solely on their acoustic signatures.

The insights gained from Phase 1 provide a solid foundation for the next steps of the project. In Phase 2, we plan to (1) mix the gunshot recordings with realistic UAV ego-noise and train noise-robust classifiers, (2) integrate deep ego-noise suppression and time-frequency masking front ends inspired by prior UAV speech enhancement work, and (3) extend the system to joint detection and sound source localization using multi-channel microphone arrays mounted on an actual UAV platform. Ultimately, the goal is to deploy a lightweight, real-time gunshot detection and localization pipeline on an embedded processor such as an NVIDIA Jetson onboard a drone, enabling rapid, safe, and autonomous acoustic situational awareness for public-safety operations.

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